Event Studies Methodology *

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1 Introduction

This lecture note gives an introduction to an important field of the empirical finance literature, the event study.[1] Event studies are an important tool in finance. Much of the corporate finance literature is concerned with the valuation of firms and the changes in firm value resulting from, for example, changes in capital structure. In general, the value of a firm is difficult to measure. However, if there is an efficient market for the firm's stock, the impact of decisions of this type can be measured by the change in the stock price around the time when the decision becomes public knowledge. Although such events can be studied in many different ways, the empirical finance literature has taken a particular approach based on statistical tests of the significance of abnormal stock returns around event dates. This type of approach also plays an important role in investment analysis. For example, one can study the performance of stocks or portfolios after an initial public offering. In this lecture note, I give an introduction and guide to this literature. The emphasis is on the methodology of conducting event studies, because an understanding of the methodology is a necessary prerequisite for understanding the literature. No attempt will be made to survey applications of event studies.

This lecture note is structured as follows. Section 2 introduces the event study methodology for events with a short event period. Section 3 discusses simulation evidence on the statistical power of the event study method. Section 4 discusses event studies with a long event horizon. Finally, Section 5 concludes with a discussion of the interpretation of the results of event studies.

[1] The event study literature has recently been reviewed by Thompson (1995); and there are some good introductions to the literature, for example Bowman (1983), Schweitzer (1988) and Peterson (1989).

2 Conducting event studies

This section gives an introduction to the empirical methodology of event studies. What exactly do we mean by an event study? To answer this question, let us consider the following well-known example: the paper by Fama, Fisher, Jensen and Roll (FFJR, 1969) that pioneered the event study methodology. We shall use this paper as an illustration throughout these notes.

FFJR consider the behaviour of stock prices around stock splits. They ask the following question: do stock prices behave differently around stock splits than in normal periods? To address this issue, they compare the holding returns on the stock around the event date (i.e. the actual date of the stock split) and the expected return if there had been no event. The difference between the actual return in the event period and the expected returns is referred to as the abnormal return. To study the effects of, in this case, stock splits the observed return series of one single firm is not very informative because returns are stochastic. Therefore, they aggregate abnormal returns over all stock splits in the sample. Statistical tests are then invoked to test the hypothesis that on average, returns around the event date are not different from their expected returns.

Of course, there are many other interesting events besides stock splits that can be studied. Bowman (1983) identifies 5 steps in conducting an event study, which I shall reduce to three:

1. Identify the event of interest and in particular the timing of the event.
2. Specify a "benchmark" model for normal stock return behaviour.
3. Calculate and analyse abnormal returns around the event date.

The following sections discuss these three stages in more detail.

2.1 Identifying the event

The type of event studied is usually motivated by economic theory. For example, consider the ex-dividend day behaviour of stock prices. The simplest efficient markets model predicts that the decline in price is equal to the dividend paid. However, there are numerous studies that find price changes significantly smaller than the dividend. In this example, the timing of the event is trivial, as the ex-dividend day is known. However, in many event studies the timing of the event is more problematic. Consider, for example, the stock price behaviour of a target firm in a corporate takeover. Identifying the actual date of the takeover as the event date would not yield meaningful results, as the takeover is usually announced a long time before and potential changes in the value of the target and bidder firms should already be reflected in the stock price. It is much more interesting to see what happens on the day that the takeover plans become public knowledge (the *announcement day*) or before that, to form an idea of the profits from insider trading. A common procedure is to pick the date of the first announcement in the Wall Street Journal, or another financial news source, of the takeover bid as the event date. Some uncertainty regarding the event date is often unavoidable, and one has to take some care in interpreting the results of an event study in such cases.

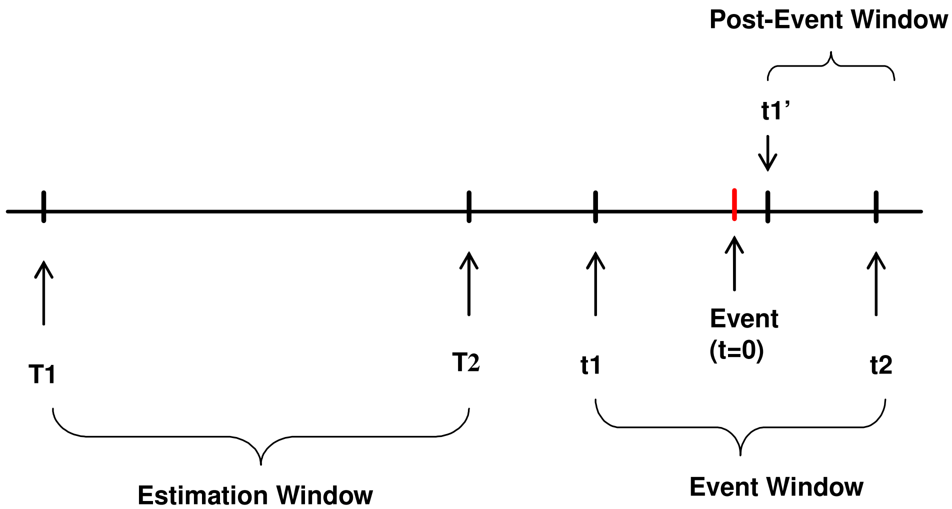
2.2 Models for the abnormal returns

An important step in conducting an event study is the choice of a benchmark model for stock return behaviour. There is a wide variety of models available in the literature, which I try to summarize here. The main differences among the models are the chosen benchmark return model and its estimation interval.

Abnormal returns (AR) are defined as the return (R) minus a benchmark or *normal* return (NR)

$$AR_{it} = R_{it} - NR_{it} \tag{1}$$

For several methods, the determination of the normal returns requires estimation of some parameters. This estimation is typically performed over an *estimation period*, $[T_1, T2]$, which precedes the event period, $[t_1, t_2]$. The event period is



typically indicated by $t = 0$. Notice that the time index t counts "event time", i.e. the number of periods (days, months) from the event and not the usual calendar time. Graphically, the time line around the event then looks like this:

2.2.1 Mean-adjusted returns

In the mean-adjusted model, the benchmark is the average return over some period, say between T_1 and T_2. The normal returns are then defined as

$$NR_{it} = \frac{1}{T} \sum_{s=T_1}^{T_2} R_{is} \tag{2}$$

where $T = T_2 - T_1 + 1$ equals the number of time periods used to calculate the average return (the length of the estimation period). The choice of the benchmark period is rather arbitrary. In an influential study, Brown and Warner (1980) use a 35-month period that ends 10 months before the event. Copeland and Mayers (1982) advocate the use of a period after the event. Many other choices are found in the literature.

2.2.2 Market adjusted returns

An obvious disadvantage of the mean-adjusted method is the omission of marketwide stock price movements from the benchmark return. Especially if the events for different firms occur at the same point in time (e.g. option expiration), the results might be biased if the whole market goes up or down in the event period. In such a case, significant abnormal returns will be detected, which may not be due to the event but rather to marketwide price movements. To correct for this omission, the return on a market index, R_{mt}, can be chosen as the benchmark:

$$NR_{it} = R_{mt} \tag{3}$$

The resulting abnormal returns are referred to as market adjusted returns. A point of interest is which market index to choose. The standard choice for US research are the CRSP equally weighted and value weighted indexes.[2] Brown and Warner (1980) compare an equally weighted and a value weighted index, and conclude that they give similar results.

2.2.3 Market model residuals and CAPM

The market adjusted returns method implicitly assumes that the "beta" of each stock is equal to one. This is obviously not always the case, and it is better to account for differences in "beta" in defining abnormal returns. Therefore, a good way to define abnormal returns is as residuals of the market model:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \epsilon_{it} \tag{4}$$

The abnormal returns are then defined as the residuals or prediction errors of this model,

$$NR_{it} = \widehat{\alpha}_i + \widehat{\beta}_i R_{mt} \tag{5}$$

where $\widehat{\alpha}$ and $\widehat{\beta}$ are OLS estimates of the regression coefficients. The period over which the market model is estimated differs among studies, but most studies use an estimation period preceding the event period or surrounding (but not including) the event period.

[2] Both these indexes are available in WRDS (the Wharton Research Data System), for which Tilburg University has a licence.

FFJR use a sample of monthly stock returns starting in 1926 and running to 1959. The parameters of the market model, (4), are estimated using the whole sample, but the period running from 15 months before the split to 15 months after the split is excluded. Note that this concerns a different calendar time period for each firm, as the events do not occur simultaneously (except by coincidence). They use the return on the equally weighted CRSP index as the benchmark market return.

An alternative to the market model is a CAPM type model, where excess returns are modeled as:

$$R_{it} - R_{ft} = \beta_i \left(R_{mt} - R_{ft}\right) + \epsilon_{it} \tag{6}$$

The associated normal return of the CAPM is

$$NR_{it} = R_{ft} + \widehat{\beta}_i \left(R_{mt} - R_{ft}\right) \tag{7}$$

Calendar time effects are a potential problem with the market model. It is well documented that returns on Monday are lower, and on Friday slightly higher, than on other trading days. If events are clustered on one of these days, the usual abnormal returns may be biased. A simple solution for this problem is to include calendar (i.e. day-of-the-week) dummies in the market model from which abnormal returns are computed (de Jong, Kemna and Kloek, 1992).

2.3 Analysing abnormal returns

In analysing abnormal returns, it is conventional to label the event date as time $t = 0$. Hence, from now on $AR_{i,0}$ denotes the abnormal return on the event date and, for example, $AR_{i,t}$ denotes the abnormal return t periods after the event. If there is more than one event relating to one firm or stock price series, they are treated as if they concern separate firms. We typically consider an event period, running from t_1 to t_2. Assuming there are N firms in the sample, we can construct a matrix of abnormal returns of the following form:

$$\begin{pmatrix} AR_{1,t_1} & .. & AR_{N,t_1} \\ | & .. & | \\ AR_{1,-1} & .. & AR_{N,-1} \\ AR_{1,0} & .. & AR_{N,0} \\ AR_{1,1} & .. & AR_{N,1} \\ | & .. & | \\ AR_{1,t_2} & .. & AR_{N,t_2} \end{pmatrix}$$

Each column of this matrix is a time series of abnormal returns for firm i, where the time index t is counted from the event date. Each row is a cross section of abnormal returns for time period t.

In order to study stock price changes around events, each firm's return data could be analysed separately. However, this is not very informative because a lot of stock price movements are caused by information unrelated to the event under study. The informativeness of the analysis is greatly improved by averaging the information over a number of firms. Typically, the unweighted cross-sectional average of abnormal returns in period t is considered:

$$AAR_t = \frac{1}{N} \sum_{i=1}^{N} AR_{it} \tag{8}$$

Large deviations of the average abnormal returns from zero indicate abnormal performance. Because these abnormal returns are all centered around one particular event, the average should reflect the effect of that particular event. All other information, unrelated to the event, should cancel out on average.

Often, we are interested not only in performance at the event date, but also over longer periods surrounding the event. The usual way to study performance over longer intervals is by means of cumulative abnormal returns, where the abnormal returns are aggregated from the start of the event period, t_1, up to time t_2, as follows:

$$CAR_i = AR_{i,t_1} \ + \ .. + \ AR_{i,t_2} = \sum_{t=t_1}^{t_2} AR_{it} \tag{9}$$

Again, in event studies the CARs are aggregated over the cross-section of events to obtain cumulative average abnormal returns ($CAAR$):

$$CAAR = \frac{1}{N}\sum_{i=1}^{N} CAR_i \tag{10}$$

Note that the $CAAR$ can also be obtained by aggregating the AAR_t's over time, as is easily checked from (8) and (9):

$$CAAR = \sum_{t=t_1}^{t_2} AAR_t$$

An example of a graph of CAR's is given in Figure 1 from Vermaelen (1981). This graphs plots the average stock price reaction in a period from 60 days before a stock repurchase to 60 days after the event. Hence, in our timing convention. the start of the event period is $t_1 = -60$ and the end, t_2, runs from -60 to 60.

FFJR calculate AARs for 29 months prior to the event through 30 months after the event. The AAR's are graphed in Figure 2a, and the CAAR's in Figure 2b. The evidence presented by these graphs is clear. In the 30 months before a stock split, the stock has performed excessively well. On average, stocks which split have had a 33% higher return than comparable stocks that did not split. However, there is no price effect of the split itself. The AAR at the event date is very small, and the CAAR remains flat after $t = 0$.

2.4 Testing abnormal performance

Although graphical reporting of cumulative abnormal returns is instructive and often suggestive, in almost all event studies the graphical analysis is supported by statistical tests. These tests are designed to answer the question whether the calculated abnormal returns are significantly different from zero at a certain, a priori specified, significance level. The null hypothesis to be tested is of the form

$$H_0 : E(AR_{it}) = 0 \tag{11}$$

Which statistical test of this hypothesis is appropriate depends on the way in which the abnormal returns are constructed and on the statistical properties of stock returns. In this section we discuss several test statistics. We pay particular attention to some pitfalls that a researcher might encounter when conducting an event study.

2.4.1 The basis: t-tests

The most common test of the null hypothesis of no abnormal return, equation (11), is a simple t-test. In order to introduce this test, we make some restrictive assumptions, but these will be relaxed later on. Specifically, assume that the abnormal returns AR_{it} that together make up the average abnormal return AAR_t are independently and identically distributed. Moreover, we assume that they follow a normal distribution with mean zero (under the null hypothesis) and variance σ^2. The independence assumption implies, for example, that all abnormal returns are cross-sectionally uncorrelated: $E(AR_{it}AR_{jt}) = 0$ for $i \neq j$. Hence, the variance of the average, AAR_t, is equal to $1/N$ times the variance of a single abnormal return, so that $AAR_t \sim N(0, \sigma^2/N)$. If σ^2 were known, a test statistic for (11) is given by

$$Z = \sqrt{N}\frac{AAR_t}{\sigma} \sim N(0,1) \tag{12}$$

Under the stated assumptions, Z follows a standard normal distribution. Of course, in practice σ is unknown. An estimator of σ can be constructed from the *cross-sectional* variance of the abnormal returns in period t:

$$s_t = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(AR_{it} - AAR_t)^2} \tag{13}$$

This yields the following test statistic for the average abnormal return

$$G = \sqrt{N}\frac{AAR_t}{s_t} \sim t_{N-1} \tag{14}$$

Under the stated assumptions, this test statistic follows a Student-t distribution with $N-1$ degrees of freedom. However, there is strong evidence that stock returns do not satisfy the normality assumption imposed to derive the distribution of the Z and G test statistics.

FFJR provide an interesting plot of the empirical distribution of the abnormal returns in their study. They graph the empirical distribution function against the normal distribution function. If the empirical distribution were normal, this graph would be approximately a straight line. However, as can be seen from Figure 1, the extreme ends of the distribution are flatter than the normal distribution. This indicates that there are more and larger extreme observations than predicted by the normal distribution. This phenomenon is observed in almost all stock return series and is generally referred to as leptokurtosis or a fat tailed distribution.

If stock returns do not follow a normal distribution, the exact small sample distribution result that G follows a Student-t distribution does no longer hold. If we maintain the assumptions that the abnormal returns are independent and have the same mean and variance, it can be shown that in large samples, G approximately follows a standard normal distribution. This is a result of the Central Limit Theorem, which states that under these assumptions, $\sqrt{N}$ times the average, divided by the standard deviation converges to a standard normal random variable

$$G = \sqrt{N}\frac{AAR_t}{s_t} \approx N(0,1) \tag{15}$$

Hence, if N is large enough, the quantiles of the normal distribution can be used as critical values for the t-test. In event studies, $N > 30$ is typically sufficient for this. For example, for a two-sided test at 5% confidence level, the critical value is 1.96. So, if $G > 1.06$ or $G < -1.96$, the null hypothesis of no abnormal performance, (11), is rejected. For a test at 10% confidence level, the critical value is 1.67, and the 1% critical value is 2.36.

2.4.2 Testing the significance of cumulative abnormal returns

Often, one is interested in the abnormal performance of event firms over a longer event period. One common situation is when the date at which the event took place cannot be determined exactly. For example, the news of a possible takeover-bid might spread gradually to the public, and may be reported with some lag in the press. If there is such event date uncertainty, the abnormal returns may be spread out around the chosen event date ($t = 0$). In such circumstances, it is often necessary to test the significance of abnormal returns in a window around $t = 0$. Also, it might be interesting to test the significance of cumulative abnormal returns over the complete pre-event or post-event period.

In this section, we describe how to test the significance of the abnormal returns over an arbitrary event interval $[t_1, t_2]$. The null hypothesis is that the expected *cumulative* price change over this period is zero. First we define the *cumulative abnormal return* as

$$CAR_i = AR_{i,t_1} + \ .. \ + AR_{i,t_2} \tag{16}$$

The null hypothesis to be tested then simply is $H_0 : E(CAR_i) = 0$. This hypothesis can be tested in a similar way as testing a one-period abnormal return. The most common procedure is the following. First, calculate the CAR_i for every event form i. Then, calculate the cross-sectional average

$$CAAR = \frac{1}{N}\sum_{i=1}^{N} CAR_i \tag{17}$$

and standard deviation

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(CAR_i - CAAR)^2} \tag{18}$$

The t-test then simply is

$$G = \sqrt{N}\frac{CAAR}{s} \approx N(0,1) \tag{19}$$

If the CAR_i's are mutually uncorrelated, this test statistic approximately has a standard normal distribution for large N.

2.4.3 Standardization

The assumption that all abnormal returns are identically distributed is usually too strong. Especially the assumption that the variance of the abnormal returns is equal for all series i (cross-sectional homoskedasticity, $\sigma_i^2 = \sigma^2$) is not likely to be true, simply because some stocks are more volatile than others. Including one or two very volatile stocks in the analysis might cause a large variation in the AAR_t and hence a low power of the test. Therefore, it seems natural to use a weighted average of abnormal returns that puts a lower weight on abnormal returns with a high variance. A frequently used weight is the time-series estimate of the standard deviation of the abnormal returns. Such an estimate can be obtained as follows. First, define the *time series* average of the abnormal returns for firm i over the estimation period $[T_1, T_2]$

$$\overline{AR}_i = \frac{1}{T-1}\sum_{t=T_1}^{T_2} AR_{it} \tag{20}$$

The time series variance for firm i is estimated by

$$s_i = \sqrt{\frac{1}{T_2 - T_1} \sum_{t=T_1}^{T_2} (AR_{it} - \overline{AR}_i)^2} \tag{21}$$

Using the estimated standard deviation s_i, we define the standardized abnormal return (SAR) as follows[3]

$$SAR_{it} = \frac{AR_{it}}{s_i} \tag{22}$$

The cross sectional average of the SAR's is denoted as follows

$$ASAR_t = \frac{1}{N} \sum_{i=1}^{N} SAR_{it} = \frac{1}{N} \sum_{i=1}^{N} \frac{AR_{it}}{s_i} \tag{23}$$

This expression makes clear that $ASAR_t$ is a weighted average of the AR's of individual firms, with weights inversely related to the estimated time-series standard deviation of that firm's abnormal return. If the variance of the AR_{it}'s is constant over the sample period, the SAR's have a variance equal to 1 (in large samples). If moreover the SAR's are uncorrelated across firms, the following test statistic can be constructed:

$$Z_s = \sqrt{N} \cdot ASAR_t = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} SAR_{it} \tag{24}$$

By the central limit theorem, this statistic approximately follows a standard normal distribution for large N. The same idea can be applied to cumulative abnormal returns.

2.5 Some problems with the tests

All tests discussed so far had the important maintained assumption that the abnormal returns were uncorrelated, both serially and cross-sectionally (although they were allowed to be cross-sectionally heteroskedastic). Moreover, it was assumed that the variance of the returns in event periods is the same as the variance in non-event periods. Both assumptions are sometimes violated, especially when dealing with daily observations. In this section we discuss four potential problems: cross-sectional dependence, event-induced variance, serial correlation, and thin or non-synchronous trading

[3] This method is often attributed to Patell (1976).

2.5.1 Cross-sectional dependence

So far, we assumed that the abnormal returns are uncorrelated between two different events, i.e. $\text{Cov}(AR_{it}, AR_{jt}) = 0$, $i \neq j$. However, if there is event clustering (several events occur in the same calendar period) there may be cross-sectional correlation between abnormal returns. In that case, the variance of the average of N abnormal returns is no longer equal to $1/N$ times the variance of a single returns, but larger (if the correlation is positive, as it typically is). As a consequence, the usual variance estimator underestimates the variance of the average abnormal returns, the usual t-statistics are biased upward, and the null hypothesis is rejected too often.

To deal with cross-sectional dependence, Brown and Warner (1980) propose the so-called *crude dependence adjustment* method. This method estimates the variance of the average abnormal returns directly from the *time series* of observations of average abnormal returns in the estimation period:

$$\bar{s} = \sqrt{\frac{1}{T-1} \sum_{t=T_1}^{T_2} (AAR_t - AR^*)^2}, \tag{25}$$

with

$$AAR_t = \frac{1}{N} \sum_{i=1}^{N} AR_{it}, \quad AR^* = \frac{1}{T} \sum_{t=T_1}^{T_2} AAR_t$$

The associated t-test statistic is

$$\overline{G} = \frac{AAR_t}{\bar{s}} \approx N(0,1) \tag{26}$$

As usual, in large samples this statistic approximately follows a standard normal distribution.

2.5.2 Event-induced variance

Another important shortcoming of all t-tests discussed so far is the assumption that the variance of the abnormal returns is the same in event and non-event periods. If there is an increase in the variance on event dates, the variances calculated over periods different from the event period underestimate the true variance. Hence, the t-test statistics will be too large in general and the null hypothesis (11) will be rejected too often.

FFJR show a plot of the mean absolute deviation of the abnormal returns around the event date. The mean absolute deviation is a measure of dispersion that is comparable to the standard deviation, but more robust against outliers. Figure 4 clearly shows that the dispersion in abnormal returns is much large at and around event dates. Hence, event induced variance is potentially a serious problem.

Boehmer et al. (1991) show that it is a robust test procedure to calculate the variance of the abnormal returns over the cross-section of events in each period. This is exactly what was already proposed in (13) and (18), so that these tests are expected to perform well even if there is event-induced variance. However, in constructing the t-tests for standardized abnormal returns, (24), we assumed that the variance of abnormal returns AR_{it} was constant over time. This assumption is violated when there is event induced variance. Boehmer et al. (1991) propose the following procedure to correct the test statistics. First, calculate SAR's in the usual way. Second, construct a t-test by dividing the average SAR's by their estimated cross sectional standard deviation:

$$\frac{ASAR_t}{S/\sqrt{N}} \tag{27}$$

with

$$S = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(SAR_{it} - ASAR_t)^2} \tag{28}$$

A similar procedure can be followed for cumulative standardised abnormal returns.

2.5.3 Serial correlation

Another potential problem is serial correlation. It is well documented that expected returns at high frequency are positively serially correlated (Lo and MacKinlay, 1988). A possible source of negative serial correlation is mentioned by Roll (1984), who stresses the point that observed transaction prices are either buying or selling prices. The bid-ask spread and the random nature of the so-called bid-ask bounce may introduce some negative serial correlation in returns. Just like in the previous two sections, positive serial correlation will lead to an underestimate of the variance of abnormal returns and to an upward bias in the test statistics. It is possible to correct the standard errors of the abnormal returns for serial correlation, for example by the method of Newey and West (1987). For short-horizon event studies, the simulation experiments of Brown and Warner (1985) show that for the levels of serial correlation found in actual data this correction is minimal. For long-horizon event studies the problem is more serious, as will be discussed in Section 4.

2.6 Conducting event studies with small cross-sections

All tests discussed so far invoke the central limit theorem to prove that their distribution under the null is standard normal. However, sometimes very small cross-sections of events are used. Especially when using daily data, the underlying abnormal returns have very fat tails. The approximation of the test statistics distributions by a normal distribution may be very poor in such small cross-sections. Typically, because stock returns are fat-tailed, the critical values of the normal distribution will be too small. Hence, one will reject the null hypothesis (11) too often. Some authors argue that even in large samples the approximation by the normal distribution is poor. FFRJ claim that stock returns have distributions which resemble so-called sum-stable distributions. For such distributions, the variance does not exist and therefore the Central Limit Theorem does not apply. As a result, the t-tests are invalid, even asymptotically.

To resolve these problems, non-parametric tests can be used. These tests are valid under very general distributional assumptions regarding the abnormal returns. Non-parametric tests may also be more robust to outliers and other data imperfections. In this section we discuss two non-parametric tests, the sign test and the rank test.

2.6.1 Sign test

The sign test tests whether there are as many positive as negative abnormal returns on event dates. The sign test statistic is based on the fraction of positive abnormal returns in the event period (denoted by p). Under the null, and if the return distribution is symmetric, the expectation of p is 0.5. The test statistic

$$Z = 2\sqrt{N}(p - 0.5) \tag{29}$$

has a standard normal distribution under the null that the *median* of the abnormal returns is zero. Therefore, the sign test in this form will only test hypothesis (11), that the *mean* of the abnormal returns is zero, if the distribution of the abnormal returns is symmetric.

Corrado and Zivney (1992) propose an adjustment to the sign test for skewed distributions. This adjusted sign test is based on the sign of $AR_{it} - M_i$, where M_i is the median of the ith abnormal return series. Defining $G_{it} = (+1, 0, -1)$ if $AR_{it} - M_i$ is positive, zero or negative, respectively, the test statistic is

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \frac{G_{it}}{s_g}, \qquad s_g = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} G_{it}^2} \tag{30}$$

If there are no G_{it}'s equal to zero, this statistic is equal to the usual sign test applied to the $AR_{it} - M_i$ series.

2.6.2 Rank test

The sign tests suffer from a common weakness: they do not take the magnitude of the abnormal return into account. In contrast, the t-tests of the previous subsection are very sensitive to the magnitude of an abnormal return. The rank test proposed by Corrado (1989) is a non-parametric way to account for the magnitude of an abnormal return, but without the distributional assumptions which are needed to make the t-tests valid. The test works as follows. Denote the rank of the abnormal return AR_{it} in the whole series (including the event period) of abnormal returns by K_{it}. This rank is transformed into the statistic $U_{it} = K_{it}/N$, which should be uniformly distributed under the null that event periods are not different from non-event periods. To test this hypothesis, define the following test statistic

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\frac{U_{it}-0.5}{s_u}, \qquad s_u=\sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(U_{it}-0.5)^2} \tag{31}$$

where 0.5 is the expected value of U_{it}. The central limit theorem can be invoked to show that the test statistic (31) follows approximately a normal distribution in large samples. Compared with the usual t-tests, the convergence to the normal distribution of the averages of the rank may be faster than the averages of the returns, especially when these have fat tails. It is therefore expected that the rank tests give better results in small cross-sections. The non-parametric tests were devised to mitigate one potential problem with the t-tests, namely, non-normality. All the other potential problems mentioned before, such as event clustering and event-induced variance, still remain. The adjustments to the tests used in that section are in most cases also applicable to the rank test. Note however that standardisation of the returns is not useful any more, as the ranking of returns remains unchanged by standardisation.

3 The performance of event studies

So far, we have discussed a number of problems that potentially show up when conducting an event study. We also have discussed a number of different test statistics adapted to these problems. It is not clear however how well these tests perform in practice and whether all problems are serious enough to warrant using complicated statistical techniques. Brown and Warner (BW 1980, 1985) have done important work to asses the performance of various test statistics used in event studies. Following their work, a number of other interesting studies have been done. In this section, we discuss the setup and report the main results from these studies.

3.1 The setup of the experiments

Without going into complete detail, we discuss here the most important features of BW's studies. Essentially, the BW papers are simulation studies, where a world is simulated where the null hypothesis (that abnormal returns on event dates are zero) is true. Instead of the usual procedure of generating artificial stock returns, they randomly draw $N = 50$ time series of specified length ($T = 60$ months or 250 days) from the population of actually observed returns on NYSE stocks. One of the last dates of each time series is, arbitrarily, defined as the event date. Then, tests for abnormal performance are calculated, given that the event date is known. The random nature of the sampling ensures that the null hypothesis is true for the simulated series. This procedure is repeated 250 times, which gives 250 artificially generated test statistics.

The performance measures considered by BW are the actual size and the power of the tests under various alternatives. The size of a test is the relative frequency of rejection of the null hypothesis when it is true, i.e. the relative frequency of a so-called type-I error. For a good test, the actual size is equal to its nominal size, i.e. the significance level chosen (usually 1%, 5% or 10%). The power of a test is the frequency of rejection of the null hypothesis when it is false. Of course, a good test rejects the null very often if it is false. In that case, the test has a high power. In order to assess the power of the tests, a fixed abnormal return (0.5% or

1% is added to the return on the chosen event date. The whole experiment is then repeated, and the power of the tests is then calculated.

3.2 The base case

BW80 compare the performance of event studies that use different benchmark models and different test statistics. The benchmark returns models used are mean-adjusted returns, market adjusted returns and market model residuals. The tests considered are the basic t-test (14), the sign test (29) and the rank test (31). BW80 report the results in their Table 1. The t-tests have the correct size, both on a 5% and on a 1% level. The rank test has an actual size which is a bit too low, while the size of the sign test is almost 0. The power of the t-test is higher than that of the rank test, which again is much higher than that of the sign test. An experiment with the same setup conducted by Corrado and Zivney (1992) shows that the rank test has the correct size and is slightly more powerful than the t-test. This is perhaps a surprising result, but note that the simulations are from the population of actual returns, which are fat-tailed. The convergence of the average of uniformly distributed rank statistics to the standard normal may be faster than the convergence of the average of the fat-tailed abnormal returns themselves. To summarize, the t-test and the rank test perform quite well, but the sign test performs badly. Therefore, it will not be considered further in this section. From Table 1 no clear choice between the benchmark return models can be made. All three models give the same performance of the test statistics.

3.3 Deviations from the base scenario

3.3.1 Event date clustering

In the base scenario, event dates were sampled randomly, so that one could safely assume that the abnormal returns were cross-sectionally uncorrelated. The effects of event clustering are studied by picking the same event date for all firms. In principle, this induces cross-sectional correlation among the abnormal returns of all firms. Two tests are compared: the simple t-test and the crude dependence adjusted t-test (26). Again, different benchmark return definitions are compared. The results in Table 6 of BW80 show that event clustering leads to a much too high actual size for the unadjusted t-test when using mean-adjusted returns. However, when using the other benchmark return definitions or the dependence adjusted t-test, there are no significant size distortions. The reason must be that the adjustment for market movements in the market adjusted and market model residuals is sufficient to remove most cross-sectional dependence among the abnormal returns. The power of the tests based on market model residuals is the highest among the benchmark models considered. Using daily data, BW85 reach essentially the same conclusions.

3.3.2 Event induced variance

BW85 induce additional variance in the event period by doubling the return on the event day. Table 9 in BW85 compares the performance of two t-tests, one that uses a time-series estimate of the standard deviation and another that uses a cross-sectional estimate. As expected, the time-series based t-test rejects the null too often if there is no effect at all, and has lower power if there is an effect. The cross-section based test performs much better on both the size and the power criterion. Boehmer et al. (1991) extend the BW85 study. Among other results, their simulations show that the rank test has the highest power, with the t-test on standardised residuals, calculated according to equation (24), as a good second.

3.3.3 Small cross-sections

The early evidence in Brown and Warner (1985) suggested that non-normality was not a problem. However, they used relatively large cross-sections of 50 events. Corrado and Zivney (1992) compare the size and power of t-tests and the rank test in small cross-sections of 10 or 20 events. The t-tests exhibit significant size distortions: there are many more Type I errors than the nominal significance level. i.e. the tests reject the null too often, even when it is true. Campbell and Wasley (1993) find similar size distortions for data from NASDAQ, even with large cross sections of 50 events. The explanation of the latter result is that the NASDAQ returns have much fatter tails than the returns used by Brown and Warner, which were sampled from more frequently traded NYSE stocks. Both studies show that the non-parametric rank test of Corrado (1989) does not have these problems. Moreover, it has a higher power than the t-tests. One of the reasons might be that the convergence of sum of the uniformly distributed rank statistics to the standard normal is faster than the convergence of fat-tailed abnormal returns.

3.3.4 Corrections for thin or non-synchronous trading

With daily data or with smaller sized firms, there may be some estimation problems with the market model, in particular due to non-trading or non-synchronous trading. It is well-known that non-trading causes the beta estimates to be downward biased. Dimson (1979) proposes a method to correct for this bias. However, the Brown and Warner (1985) study finds that using the bias-corrected beta estimates does not change the performance of the test statistics. This is perhaps not surprising because there was no noticeable difference between market adjusted returns (with implicit $\beta = 1$ assumption) and the market model residuals. Maynes and Rumsey (1993) challenge this conclusion and find that the t-tests may lead to erroneous inference, whereas the non-parametric test of Corrado (1989) is not affected by thin trading.

3.4 The power of the tests under event date uncertainty

All experiments discussed in the previous sub-section assumed that the event date was known beforehand. In many practical situations there is considerable uncertainty on the actual date of the event. For example, news of a merger may be leaked to the market a few days before the first announcement in the newspapers. In such a case, the significance of cumulative abnormal returns (CAR) over some period should be tested (cf. section 2.3.4). BW80 (their Table 4) assess the power of the t-tests on CAR's as follows. First, the abnormal returns is added to a randomly selected date in the interval from 10 months before to 10 months after the selected event date. The t-test (19) is used to test the significance of the abnormal returns over this interval. Not surprisingly, the power of the tests on CAR's is generally much lower than in the case where the event date is known with certainty. The intuition for this result is easy: the longer time span over which price changes are measured increases the variance of the abnormal returns, whereas the true abnormal return remains unchanged. Figures 1, 2 and 3 in BW80 make this point clear: the longer the length of the cumulation period, the higher the variance of the average abnormal returns. Hence, with the same sample size, in the case of event date uncertainty the tests are less powerful. Brown and Warner (1985) repeat this experiment for daily data and reach the same conclusions.

3.5 Overview of the results

In this section, we surveyed the performance of tests commonly used in event studies. The general conclusions can be summarized as follows:

- When using large cross-sections of frequently traded stocks, the t-tests and the non-parametric rank tests have the correct size and similar power. The power of the sign test is always lower than the power of the t-test and the rank test.

- When using smaller cross-sections and/or returns with very fat tails (e.g. NASDAQ returns) the t-tests show significant size distortions. In such cases, the rank test retains the correct size and has higher power.

- Of all the potential statistical problems discussed, only event-induced variance appears to be a serious problem. The solution in that case is to estimate the variance of the average abnormal returns directly over the cross-section of returns in the event period.

- Other potential problems, such as cross-sectional dependence, serial correlation and thin trading vanish if abnormal returns are defined as market model residuals.

- The power of tests decreases a lot if there is event date uncertainty.

4 Long-horizon event studies

The discussion in the preceding sections focused on events that lasted for a fairly short time period. Most corporate events, for example mergers and takeovers, take only a few months with most of the abnormal returns occurring in a few days around the announcement. There is a fairly large literature, however, that considers the long-term impact of corporate events. The prime example here is the study of equity offerings, both initial public offerings (IPO) and seasoned equity offerings (SEO). The literature in this area typically studies the returns of firms that have an IPO or SEO over a horizon of 36 to 60 months. In this section, we discuss some methodological issues in conducting such long horizon event studies. We do not review the empirical literature on long-horizon event studies. A good reference paper for the empirical IPO and SEO literature is Ritter and Welch (2002).

4.1 Defining abnormal returns

The correction for market returns is typically sufficient for short horizon event studies, but in long-horizon event studies the market model or the CAPM have several disadvantages. There are several well-known deviations from the CAPM, such as the size effect, the book-to-market effect and the momentum effect, see for example Bodie, Kane and Marcus (2005, Chapter 12). In long horizon event studies, the Fama and French (1996) three factor model is therefore often used as the benchmark model to generate normal returns. This model extends the market model with the returns on a "size" portfolio (SMB) and a "value" portfolio (HML)

$$R_{it} - R_{ft} = \alpha_i + \beta_i (R_{mt} - R_{ft}) + s_i SMB_t + h_i HML_t + \epsilon_{it} \tag{32}$$

where SMB ("small minus big") is the difference in return between a portfolio of small firms and a portfolio of large firms, and HML ("high minus low") is is the difference in return between a portfolio of firms with a high book-to-market ratio ("value" firms) and a portfolio of firms with a low book-to-market ratio ("growth" firms). For more details on the construction of these factors, see Bodie, Kane and Marcus (2005, Section 13.3). The normal return for firm i in period t then is defined by

$$NR_{it} = R_{ft} + \widehat{\beta}_i \left(R_{mt} - R_{ft}\right) + \widehat{s}_i SMB_t + \widehat{h}_i HML_t \tag{33}$$

The abnormal returns constructed from the three factor model are not only more accurate than the market model based abnormal returns, but they also show less cross-sectional correlation. IPO's are typically small growth firms, and will exhibit similar exposures to the size an value factors. Omitting these factors from the normal return benchmark model will lead to abnormal returns that are correlated across firms that have an IPO in the same month. Cross-sectional correlation is an important issue for the inference, as we shall see in the next subsection.

As an alternative to the Fama-French three factor model, Barber and Lyon (1997) advocate a non-parametric approach, where the benchmark return equals the return on a firm (or a portfolio return on a small group of firms) with similar size and book-to-market ratios. This approach is more flexible than the linear regression in (32), but it needs more data to obtain the same power. In practice, therefore, the Fama-French three factor model remains a popular benchmark in long-horizon event studies.

The formal definition of the CAR in long-horizon event studies is

$$CAR_i = \sum_{t=1}^{H} AR_{it} = \sum_{t=1}^{H} \left(R_{it} - NR_{it}\right) \tag{34}$$

where the event period runs from time $t = 0$ to $t = H$; R_{it} is the return on firm i in month t after the event; and NR_{it} the corresponding normal (or benchmark) return. The CAR methodology implicitly assumes a monthly rebalancing of the portfolio to an equal weighting of the return on event firms. In practice, however, many investors have longer holding periods than one month and do not rebalance their portfolio on a regular basis. An alternative approach to performance measurement therefore is to construct returns until the end of the event period (typically 36 or 60 months) without rebalancing. These returns are so-called buy-and-hold abnormal returns, or $BHAR$. Barber and Lyon (1997) give the formal definition for the buy-and-hold abnormal return

$$BHAR_i = \prod_{t=1}^{H} \left[1 + R_{it}\right] - \prod_{t=1}^{H} \left[1 + NR_{it}\right] \tag{35}$$

The distribution of these $BHAR$'s is much more skewed than the distribution of the CAR, because over such a long period typically a few firms have extremely high returns, whereas the majority of firms has moderate even negative returns. However, in large samples this skewness is not a problem: by the central limit theorem, t-statistics based on the $BHAR$ are also approximately normally distributed. In smaller samples, however, this skewness may create problems. Fama (1998) argues that the CAR is the most appropriate definition of outperformance, but not all researchers agree with this. In the remainder of this section, the discussion refers to CAR's, but all methods can be applied to $BHAR$'s as well.

4.2 Testing abnormal performance

The usual measure of outperformance is the equally weighted average of the cumulative abnormal return

$$CAAR_{EW} = \frac{1}{N} \sum_{i=1}^{N} CAR_i \tag{36}$$

and standard deviation

$$s_{EW} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (CAR_i - CAAR)^2} \tag{37}$$

Testing the null hypothesis of no abnormal performance can be done by the t-test defined in equation (19):

$$G_{EW} = \sqrt{N} \frac{CAAR_{EW}}{s_{EW}} \approx N(0,1) \tag{38}$$

This t-test is valid if the abnormal returns are uncorrelated.

In long horizon event studies, cross-sectional correlation due to event date clustering is an important issue. Typically, there are many events in any given month. For example, in the US, on average over the period 1960–2005, there are 25 IPO's per month. The t-statistic for the equally weighted CAR, as defined in equation (38), implicitly assumes that the events are cross-sectionally uncorrelated. If returns exhibit cross-sectional correlation, the usual t-statistic will be inflated and the test will reject the null of no abnormal performance too often. A simple, but crude, correction for this cross-sectional dependence is to group all IPO's in the same month, and then calculate the standard deviation of the CAR over all groups. First, calculate the CAR for all firms with the event in *calendar* month j

$$CAR_j = \frac{1}{N_j} \sum_{i=1}^{N_t} CAR_i \tag{39}$$

where $j = 1, .., J$ counts every calendar month in the sample, and N_j is the number of events in calendar month j. The total number of events equals $N = \sum_{j=1}^{J} N_j$. The cross-sectionally weighted average abnormal performance then is defined as

$$CAAR_{CW} = \frac{1}{J} \sum_{j=1}^{J} CAR_j \tag{40}$$

with standard deviation

$$s_{CW} = \sqrt{\frac{1}{J-1} \sum_{j=1}^{J} (CAR_j - CAAR)^2} \tag{41}$$

and t-test statistic

$$G_{CW} = \sqrt{J} \frac{CAAR_{CW}}{s_{CW}} \tag{42}$$

This test statistic is robust against cross-sectional correlation, but may have less power than the usual t-test if the cross-sectional correlation is (close to) zero, see Loughran and Ritter (2000).

4.2.1 Serial correlation

Another, and often overlooked, problem in long-horizon event studies is serial correlation. This means that CAR's from events in different months can be correlated, because the event period is long and overlaps with the returns of event forms in neighboring months. This causes the CAR's (and also the month-by-month grouped CAR's) to be potentially correlated. This correlation leads to inflated t-statistics, as both the equally weighted and the cross-sectionally weighted t-tests, (38) and (42), assume that event returns in different months are uncorrelated. Mitchell and Stafford (2000) show that cross-sectional correlation between the event returns implies serial correlation of event returns, and that this seriously affects the inference. They show that the t-statistic will be inflated by a factor

$\sqrt{1+(N-1)\rho}$, where ρ is the cross-sectional correlation between event returns, and N is the number of events in the sample. Even if the cross-sectional correlation is small, the impact on the t-test may be substantial. Mitchell and Stafford (2000) estimate the cross-sectional correlation to be $\rho = 0.02576$, which is small, but with $N = 4,439$ (the number of IPO events in the US) this implies that $\sqrt{1+(N-1)\rho)} \approx 4$, so the usual t-statistics have to be divided by 4 to get the proper inference.

A correct procedure for the inference with possibly serially correlated returns is to calculate standard errors by the procedure of Newey and West (1987). This method corrects the standard errors for possible serial correlation, as follows

$$s_{NW} = \sqrt{\frac{1}{J}\sum_{j=1}^{J}\left[\widetilde{CAR}_j^2 + \sum_{k=-H+1}^{H-1} w_k \widetilde{CAR}_j \widetilde{CAR}_{j+k}\right]} \tag{43}$$

with $\widetilde{CAR}_j = CAR_j - CAAR$ and $w_k = \frac{H-|k|}{H}$. The t-test can then be calculated in the usual way.

4.3 Calendar time abnormal returns

A third way to test the significance of long-horizon event returns is the "calendar time returns" approach of Jaffe (1974) and Mandelker (1974) and advocated by Fama (1998). This approach first constructs a time series of portfolio returns, where for each month the portfolio consists of all firms that had an IPO in the last H months (typically, $H = 36$ or $H = 60$). If there is a month where no single firm had an IPO in the previous H months, the portfolio return is put equal to the risk free return. This procedure gives a monthly time-series of event portfolio returns, denoted by R_{pt}. This return is then regressed on the three Fama-French factors

$$R_{pt} - R_{ft} = \alpha_p + \beta_p (R_{mt} - R_{ft}) + s_p SMB_t + h_p HML_t + \epsilon_{pt} \tag{44}$$

The intercept of this regression, α_p, measures the abnormal performance with respect to the three factor benchmark. The significance of the abnormal performance can be tested by the t-test for the significance of α_p in this regression. The advantage of the calendar time returns approach over the usual event study method, which uses event-time CAR's or $BHAR$'s, is that cross-sectional and serial correlation are not a problem.

5 Interpretation and conclusions

In these lecture notes, we provided an introduction to the methodology of event studies. We discussed the various parametric and non-parametric tests which are popular in the applied literature. Indeed, the statistical methods used in conducting event studies are often complicated and confusing. Nevertheless, most extensions of the basic t-tests are designed to deal with serious empirical problems. Neglecting problems such as cross-sectional dependence and event-induce variance may easily lead to spurious inference. However, there is also much to gain from a careful selection of the data and an exact determination of the event dates. To quote Brown and Warner (1980), "a good use of time is still in reading old issues of the Wall Street Journal to more accurately determining event dates". The interpretation of the results of event studies is not always easy. For example, the work on mergers and takeovers leaves us with an unexplained decline of stock prices after the merger process has been completed. Skeptics of the efficient markets hypothesis will be tempted to interpret such results as a violation of market efficiency. On the other hand, Fama (1991) argues that event studies provide the cleanest evidence we have on market efficiency, and that this evidence is largely favorable to the EMH. For example, the work on stock repurchase announcements shows that the new information contained in such announcements is fully and quickly reflected in stock prices.

Long horizon event studies pose some additional problems. The measurement of the abnormal returns needs to be done more carefully, using either a multifactor benchmark model, or a matched firm approach. The definition of abnormal performance using either cumulative abnormal returns or buy-and-hold abnormal returns is also an issue. Inference in long horizon event studies is more difficult than in short horizon studies, as cross-section and serial correlation can seriously distort the t-tests. The paper by Ritter and Welch (2002) gives an excellent overview of these issues, as well as the empirical results concerning the abnormal performance of equity issuing firms.

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Appendix: Figures and Tables

Figure 1 from Vermaelen (1981).

Figures 1, 2 and 4 from Fama, Fisher, Jensen and Roll (1969).

Tables and figures from Brown and Warner (1980).

Tables from Brown and Warner (1985).

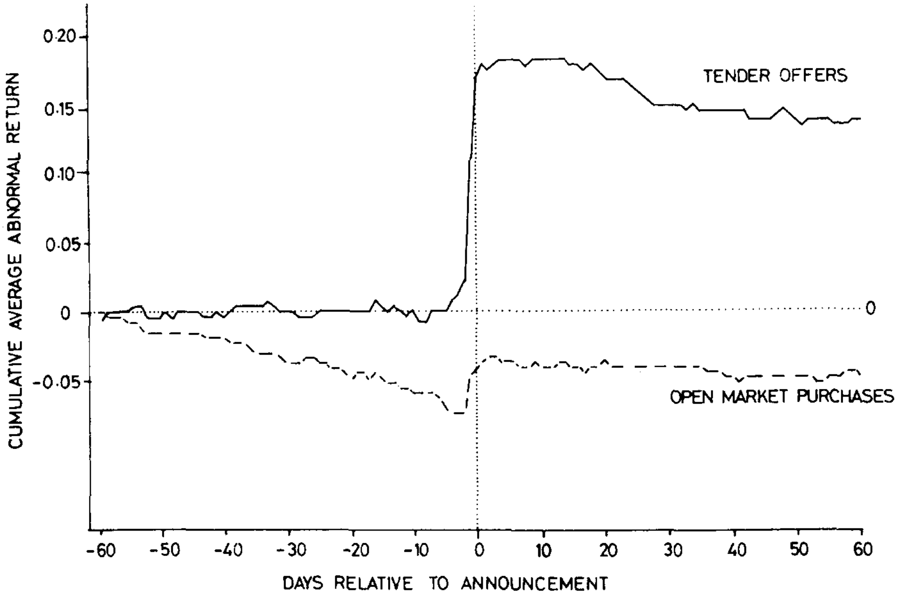

Fig. 1. Cumulative average abnormal return 60 days before the announcement until 60 days after the announcement of a sample of 131 repurchases via tender offer and 243 open market purchases.

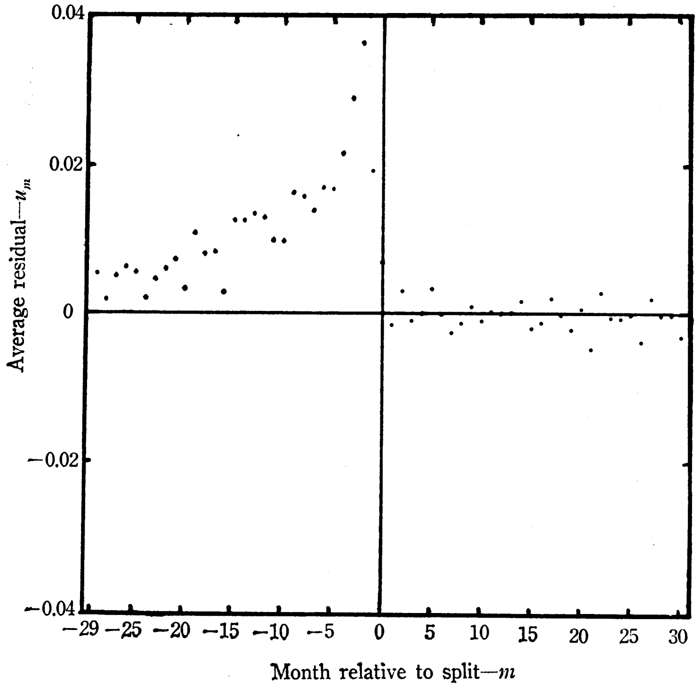

FIGURE 2a

AVERAGE RESIDUALS—ALL SPLITS

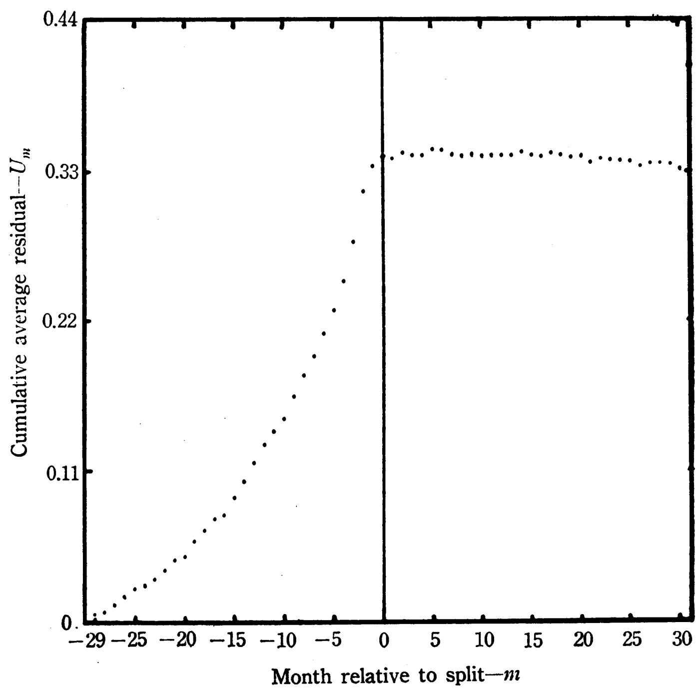

FIGURE 2b

CUMULATIVE AVERAGE RESIDUALS—ALL SPLITS

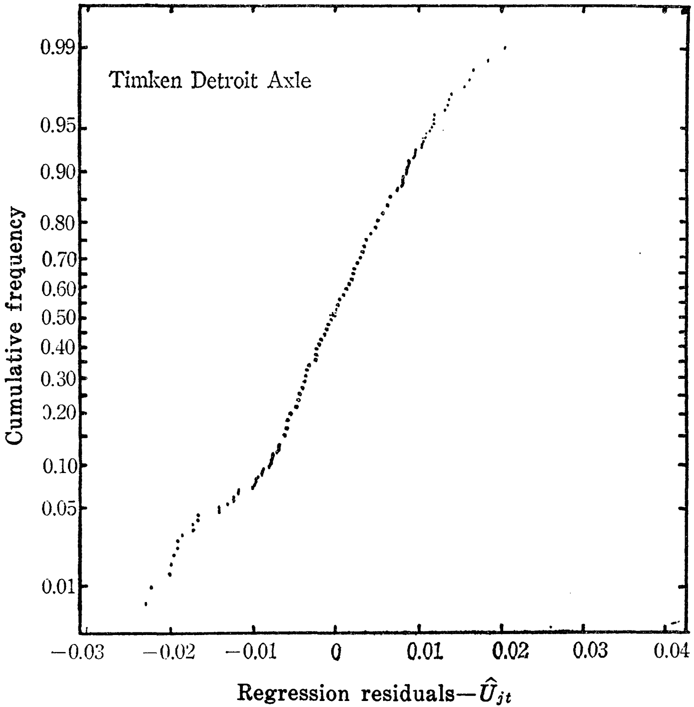

FIGURE 1

NORMAL PROBABILITY PLOT OF RESIDUALS*

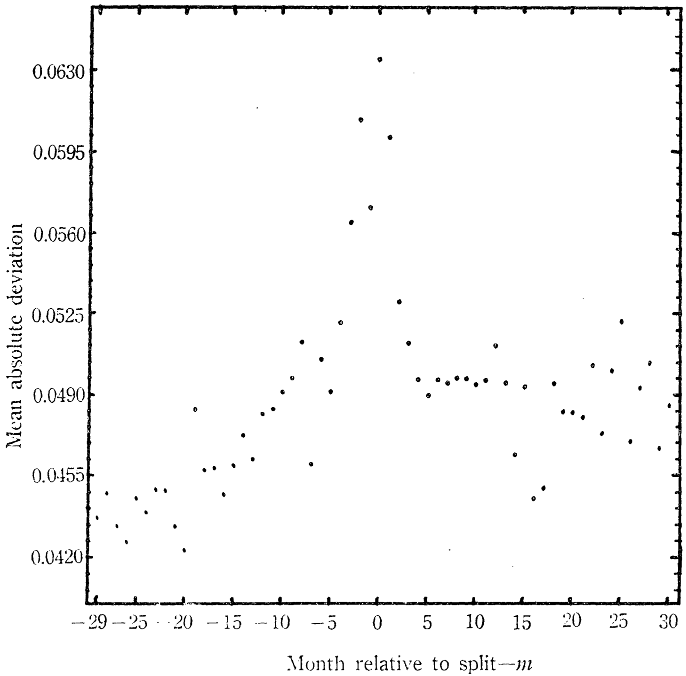

FIGURE 4

CROSS SECTIONAL MEAN ABSOLUTE DEVIATION OF RESIDUALS—ALL SPLITS

Table 1

A comparison of alternative performance measures. Percentage of 250 replications where the null hypothesis is rejected. One-tailed test. H_0: mean abnormal performance in month '0' = 0.0. Sample size = 50 securities.

Method	Test level: $\alpha = 0.05$			Test level: $\alpha = 0.01$		
	Actual level of abnormal performance in month '0'					
	0%	1%	5%	0%	1%	5%
Mean Adjusted Returns						
t-test	4.0	26.0	100.0	1.6	8.8	99.2
Sign test	0.8	6.4	96.0	0.0	1.6	90.8
Wilcoxon signed rank test	1.6	12.8	99.6	0.4	4.4	97.6
Market[a] Adjusted Returns						
t-test	3.2	19.6	100.0	1.6	5.2	96.4
Sign test	0.0	9.2	99.2	0.0	2.0	97.6
Wilcoxon signed rank test	1.6	17.2	99.6	0.4	4.4	98.8
Market[a] and Risk Adjusted Returns						
t-test	4.4	22.8	100.0	1.2	6.8	98.4
Sign test	0.4	7.2	99.6	0.0	2.4	98.0
Wilcoxon signed rank test	2.8	16.4	100.0	0.0	4.4	99.2

[a]Fisher Equally Weighted Index. Note that for 15 and 50% levels of abnormal performance, the percentage of rejections is 100% for all methods.

Table 6

The effect of event-month clustering.[a] Percentage of 250 replications where the null hypothesis is rejected ($\alpha = 0.05$). One-tailed t-test results. H_0: mean abnormal performance in month '0' = 0.0.

Method	Actual level of abnormal performance in month '0'		
	0%	1%	5%
Mean Adjusted Returns			
No Dependence Adjustment	32.4	44.4	74.0
Crude Adjustment	3.2	4.8	31.6
Jaffe–Mandelker	3.6	5.6	34.4
Market Adjusted Returns			
No Dependence Adjustment	3.6	22.8	99.6
Crude Adjustment	4.0	23.6	99.6
Jaffe–Mandelker	5.2	24.4	99.6
Market Model Residuals			
No Dependence Adjustment	4.0	23.2	99.2
Crude Adjustment	5.6	24.8	99.6
Jaffe–Mandelker	6.0	26.8	99.6
Fama–MacBeth Residuals			
No Dependence Adjustment	4.0	25.2	100.0
Crude Adjustment	4.8	24.4	98.8
Jaffe–Mandelker	4.8	26.4	99.2
Control Portfolio			
Crude Adjustment	4.4	23.2	99.2

[a]For a given replication, month '0' falls on the same calendar date for each security. The calendar date differs from replication to replication. Equally Weighted Index.

[b]Methodology not readily adapted to other dependence adjustment procedures.

Table 4

Alternative performance measures when the precise date of the abnormal performance is unknown.[a] Percentage of 250 replications where the null hypothesis is rejected ($\alpha = 0.05$). One-tailed t-test results using Equally Weighted Index. H_0: mean abnormal performance in the interval $(-10, +10) = 0.0$

	Actual level of abnormal performance in interval $(-10, +10)$				
Method	0%	1%	5%	15%	50%
Mean Adjusted Returns	7.6 (9.2)	9.2 (13.6)	28.4 (35.2)	82.0 (94.4)	100.0 (100.0)
Market Adjusted Returns	3.6 (5.2)	5.6 (6.8)	18.4 (35.2)	73.2 (96.4)	100.0 (100.0)
Market Model Residuals	7.2 (7.6)	10.8 (10.4)	24.4 (39.6)	86.4 (96.8)	100.0 (100.0)
Fama–MacBeth Residuals	3.6 (8.8)	5.2 (15.6)	16.4 (45.6)	74.8 (97.6)	100.0 (100.0)
Control Portfolio	4.8 (5.6)	6.4 (6.8)	16.0 (32.0)	70.0 (91.2)	100.0 (100.0)

[a]For each security, abnormal performance is introduced for one month in the interval $(-10, +10)$ with each month having an equal probability of being selected. The rejection rates shown in brackets are for the case where (1) for each security, abnormal performance is introduced for one month in the $(-5, +5)$ interval, with each month having an equal probability of being selected, and (2) the null hypothesis is that the mean abnormal performance in the $(-5, +5)$ interval is equal to 0.

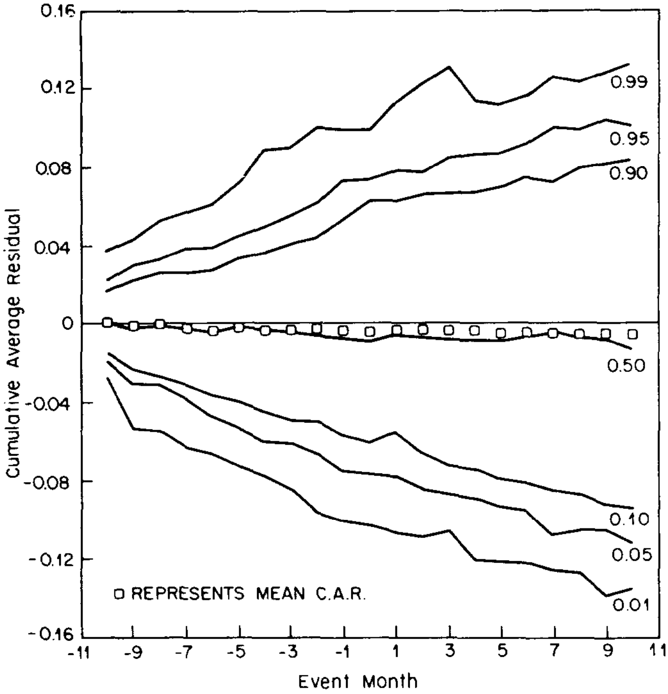

Fig. 1. Fractiles of cumulative average residual under null hypothesis of no abnormal performance.

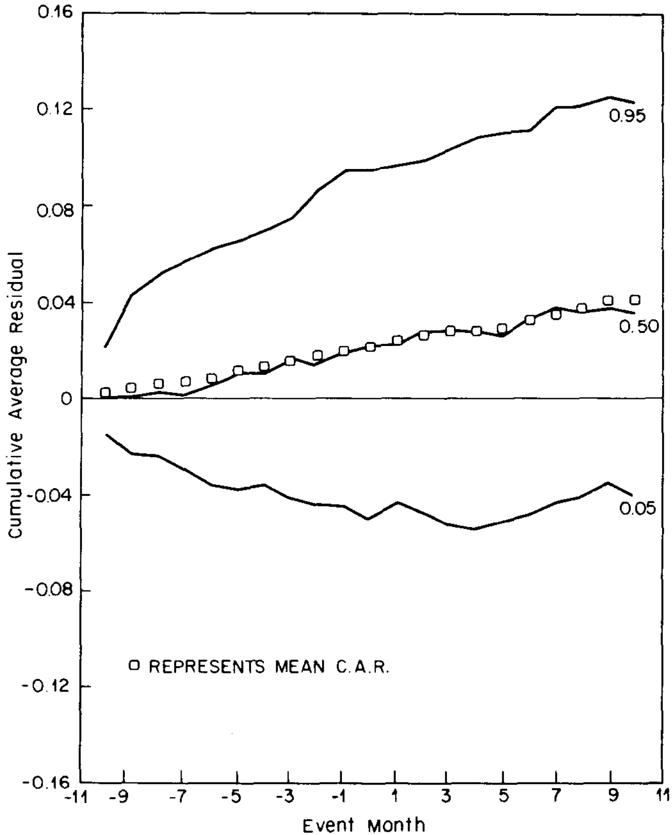

Fig. 2. Fractiles of cumulative average residual with 5% excess return distributed uniformly on months −10 to +10.

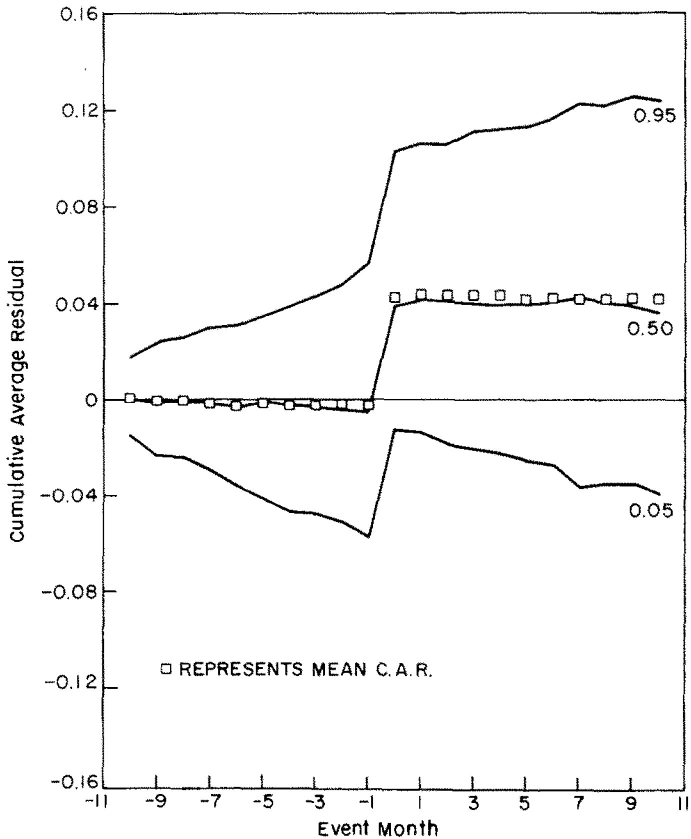

Fig. 3. Fractiles of cumulative average residual with 5% excess return in month zero.

Table 9

The effect of variance increases during the event period. For each security, the day '0' variance is doubled (see text). Percentage of 250 samples where the null hypothesis is rejected; market model. H_0: mean abnormal performance at day '0' = 0.[a]

Variance estimation method		Actual level of abnormal performance at day 0: 0	0.01
Use time-series data from estimation period	Variance increase	12.0%	70.4%
	No increase	4.4	80.4
Use cross-sectional data from event period	Variance increase	4.0	48.8
	No increase	3.6	80.4

[a]Sample size = 50; one-tailed test, $\alpha = 0.05$; randomly selected securities and event dates 1962–1979.